Visualizing Multivariate Time Series Data in a Biological Input and Output System

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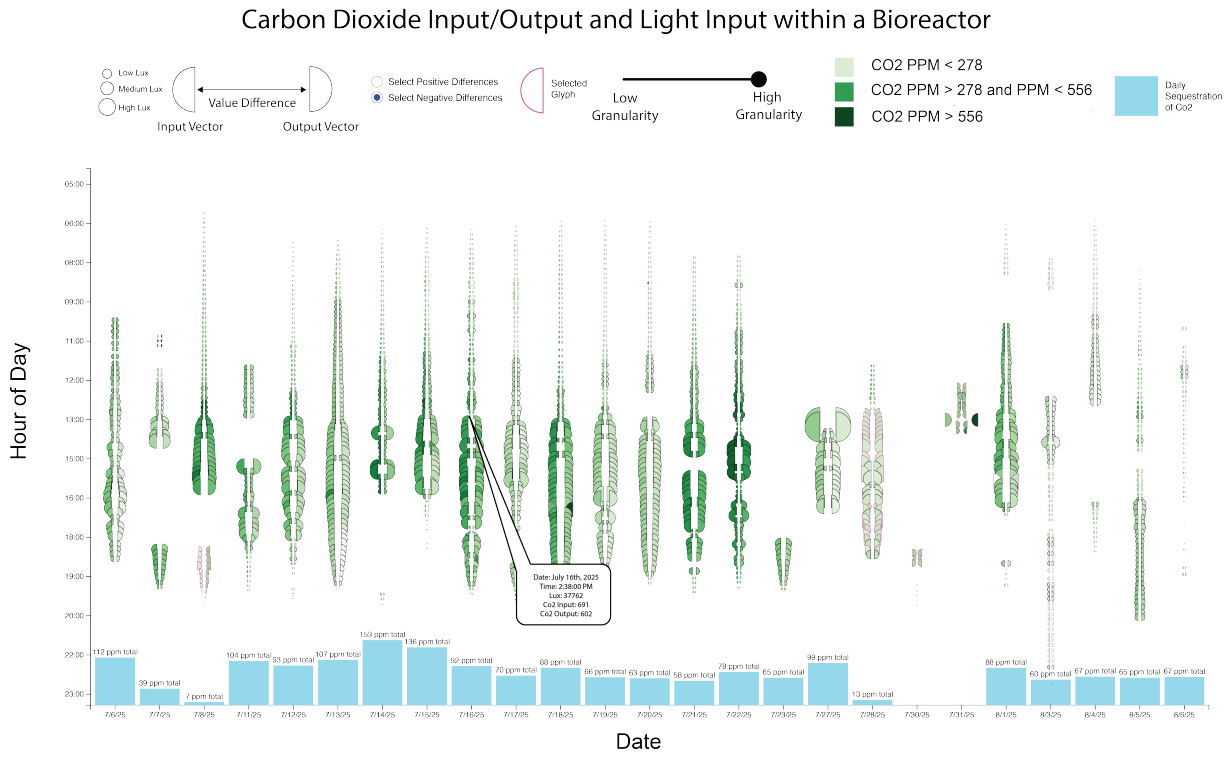


Figure 1: CO_2 input/output data and light data throughout July 2025 from bioreactors. The size of the semicircles encodes light values, while the color indicates concentration of CO_2. The distance between the two semicircles represents the difference in CO_2 input and output, with negative differences highlighted in magenta. The height of the bar at the bottom encodes daily aggregated CO_2 sequestration. The figure reveals the bioreactor sequesters the most CO_2 in the afternoons when sunlight is most intense.

Abstract

As architects increasingly explore the usage of biological systems in sustainable building practices, performance validation becomes a critical need of architects and adopters of this technology. Despite the existing research in temporal multivariate visualizations related to natural systems, there exists a need for a visualization that allows users to draw quick insights from a specific configuration. We set forth a novel approach in the visualization of natural systems that are integrated into architectural facades. In this display, we introduce a scalable set of graphs that model multivariate data over daily cycles. We allow users to discover trends over time, detect subtle changes, and draw conclusions regarding system effectiveness. Our display employs glyphs that encode multiple variables into one data point, whilst allowing for a natural discovery of relationships between variables. We finally discuss how interaction design and scalability can be further improved in future iterations of this system.

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1 Background

Time series visualizations are often limited by the number of variables they can display. Many different variables displayed on one graph are prone to cause visual clutter and increase cognitive effort required to decode them [4]. We especially observe this problem unfold in environmentally driven events as there are often multiple independent variables acting on a dependent variable over extended periods of time [3]. Bioreactors that contain microalgae are a biological system that can improve indoor air quality by sequestering carbon, measured as the difference between carbon dioxide (CO_2) input and output. Their effectiveness depends on solar exposure throughout seasons and other environmental factors. [2, 6]. Sustainable architects and building managers seek a visualization system to: (1) Optimize performance by analyzing carbon reduction time series in relation to environmental inputs (i.e. light, (CO_2, etc.), and (2) Engage the public by showcasing how the system reduces indoor air pollution through sequestration of (CO_2).

The challenge of these tasks comes when looking at how we can model input variables and their respective outputs without overwhelming the user. Figure 1 is a visualization that demonstrates the interaction of independent environmental variables in an enclosed bioreactor system. This novel visualization allows users to accomplish the following tasks: (1) Visualize key variables in a time series visualization and (2) Perform correlation and trend analysis.

Our data were collected from a bioreactor integrated into an architectural facade Figure 2 created by Dr. Kyoung-Hee Kim. The sunlight received by the bioreactor was measured in lux using a light sensor attached to its surface. CO_2 input and output were measured in ppm (parts per million) by two sets of CO_2 sensors [5]. Efficiency of the system in sequestering CO_2 is calculated by: $CO_{2_{in}} - CO_{2_{out}}$. The larger the positive difference, the more efficient the system is in sequestering CO_2.



Figure 2: A row of bioreactors in a southwest orientation, which are growing microalgae.

2 Requirements and Design

Our key design requirements in this visualization included the following: scalability, subtle change detection, correlation exploration, and public friendliness. In representing data over time, it was crucial that we featured both diurnal cycles (represented by hours) as well as expansive seasonal cycles (represented by days), so that patterns could be derived from different levels of chronological granularity. We took inspiration from how Baur displayed music listening data in relation to time, using both axes as representations of time [1]. A singular day could then be displayed over periods of weeks to show trends over longer periods of time.

Correlation exploration factors into the time representation as users can observe how sunlight, an environmental variable dependent on time, interacts with other input variables to affect the overall system output. This is especially noticeable during the midafternoon, when bioreactors would receive maximum amount of light due to their southwest orientation. The subtle changes between CO_2 inputs and outputs can be observed within diurnal cycles, helping to tell the story of how bioreactor production peaks at certain times. This informed our design process as we sought a visualization that emphasized minor shifts in light values or CO_2 inputs/outputs.

In our design, we used color as a representation for CO_2 concentration with darker colors representing higher concentrations. We also decided on using glyphs for representing CO_2 input and output, as semicircles would naturally represent two parts of a "whole" when put together. In this instance the left semicircle represents input, the right semicircle represents output, and the sequestered CO_2 is represented by the distance in between the two semicircles. Positive differences (CO_2 output < CO_2 input) indicate that the system is functioning as expected and is sequestering CO_2. A negative difference indicates the opposite. We use the inverse hyperbolic sine function, $\text{arcsinh}(x)$, to model CO_2 change over time, so that our data is normalized logarithmically. An alternate design mapped the data points in a linear fashion, which might be useful in showing more precise differences in data over short bursts of time, but can easily cause data points to overflow across the x-axis in larger datasets due to outliers. We aggregated our carbon sequestration differences every day to offer a quick way for users to gain insights about sequestration across a longer period. Finally, we allow users to adjust the frequency of sampled data (e.g., 1 sample every 30 minutes vs. 1 sample every minute) with a slider (see Figure 3), and they can choose to highlight either positive or negative differences.

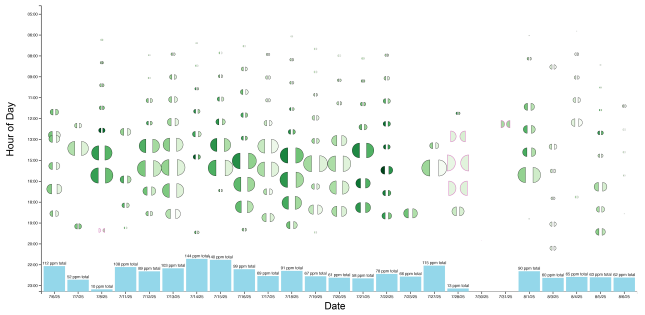


Figure 3: The same graph as in Figure 1, except with less data points therefore signaling less granularity.

3 User Study, Discussion, and Future Work

Throughout our design process we received feedback from an architect and building manager, both of whom are within the intended audience for this visualization. They both agreed that colors and the distance metaphor for representing the difference between CO_2 input and output were useful in observing general correlation. Additionally, the aggregated data at the bottom was useful for the architect to quickly observe what days had the highest sequestration. Both users commented that interpreting the glyphs was a bit tricky at the beginning; a clearer way of introducing them would have helped.

Scalability is a major consideration in future revision of our visualization. The current visualization can only show data from one set of bioreactors. Users have the need to compare data from bioreactors with different solar orientations (e.g. SE, NW, etc.) or at different locations. Future work will address this concern by allowing users to compare multiple systems in different configurations over time. Our hope is that this display extends beyond bioreactors, into other integrated natural systems that are put to a practical use (i.e. architecture). When addressing scalability in relation to time, aggregating data points and then displaying them across a long timeline with zooming features, is an avenue by which we can continue to display trends, without so much noise when expanded across a time span of months. Another solution will be to place these graphs on multiple rows, somewhat like a calendar.

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